

Eranga Kavisanka (He/Him)

Software Engineering Undergraduate | Freelancer



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OVERVIEW

Software Engineering undergraduate specializing in **DevOps**, **Cloud Computing**, and **Site Reliability Engineering (SRE)** with industry-recognized certifications including the **AWS Certified Cloud Practitioner (AWS CCP)** and **Kubernetes and Cloud Native Associate (KCNA)**. Hands-on experience with **AWS**, **Kubernetes**, **Docker**, **Terraform**, **Ansible**, **CI/CD**, **Infrastructure as Code**, and **DevSecOps**. **Batch Top**, **published researcher**, and **open-source contributor** passionate about building scalable, reliable, and cloud-native systems.

EDUCATION

B.Comp. (Hons) in Software Engineering (UG), 2023 – present
University of Sri Jayewardenepura

GPA: **3.89/4**, **Batch Top** in Software Engineering Degree Program

Expertise: *Cloud Computing, Object-Oriented programming, Software Engineering, Programming Languages, Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Deep Learning, Computer Networks*

GCE Advanced Level Examination (Physical Science), *Pinnawala Central College - Kegalle* 2021

OPEN SOURCE CONTRIBUTIONS

WSO2 Open Source Contributor – Contributed to the WSO2 Docs Integrator project with a merged pull request improving technical documentation [\[CONTRIBUTION\]](#)

MAYHEM Kubernetes Incident Simulator – Contributed to testing and validation, providing technical feedback; received a recommendation from the project's DevOps & SRE Tech Lead. [\[RECOMMENDATION\]](#)

PROFESSIONAL EXPERIENCE

Co-Founder & Full Stack Software Developer, InventX Solutions 2024 – Present

Co-founded a software startup delivering production-ready web and mobile applications for clients, participating in requirements analysis, system design, development, deployment, and maintenance.

- *Sekin Express Delivery Platform* - Delivered a production-ready logistics management platform for a real-world client, enabling secure multi-role operations, parcel tracking, and centralized delivery management. [\[Live\]](#)
- *QS Monach Corporate Website* — Responsive corporate website with a modern user interface and performance optimization. [\[Live\]](#)
- *SkillFinder Mobile Application* — Flutter mobile application connecting users with skilled professionals with an integrated administrative portal. [\[GitHub\]](#)

Banking Intern, People's Bank October 2022 – June 2023

Supported daily branch operations, customer service, and administrative banking processes.

RESEARCH PUBLICATIONS

A Review and Comparative Analysis of RF-Based Offline Communication Systems and IoT Integration Approaches for Traveller and Hiker Safety

Jayewardenepura Computing Symposium (JCoS) 2026

Reviewed and compared existing RF-based communication technologies, mesh networking protocols, and IoT architectures, identifying challenges and future directions for resilient off-grid emergency communication systems.

CERTIFICATIONS

AWS Certified Cloud Practitioner (AWS CCP)	<i>Amazon Web Services</i>
Kubernetes and Cloud Native Associate (KCNA)	<i>Linux Foundation</i>
Aviatrix Certified Multicloud Network Associate (ACE)	<i>Aviatrix</i>
Linux Foundation Training- LFS101, LFS158, LFS162, LFS167, LFS169	<i>Linux Foundation</i>
Full Stack Web Development	<i>Skyrek</i>

SELECTED PROJECTS AND EXPERIENCES

Hybrid RF-Based Off-Grid Mesh Communication Network with Opportunistic Cloud Synchronization (Ongoing Final Year Project)

Tech Stack: ESP32-C3, IoT, AWS IoT Core, MQTT, Grafana, Flutter, MySQL

Developing a hybrid RF mesh communication system enabling offline emergency messaging, opportunistic cloud synchronization, and real-time IoT monitoring. Integrating 6+ cloud and edge components including AWS IoT Core, MQTT, local persistence, and Grafana dashboards while developing expertise in distributed IoT systems, embedded communication, and cloud-connected edge architectures. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

Production-Grade GitOps Platform on Amazon EKS

Tech Stack: Amazon EKS, Terraform, GitHub Actions, Argo CD, Helm, Docker, Amazon ECR, SonarQube

Engineered a GitOps platform spanning 3 repositories (Application, GitOps, Infrastructure) with automated infrastructure provisioning, CI/CD, quality gates, container delivery, and continuous reconciliation. Improved proficiency in GitOps workflows, Infrastructure as Code, Kubernetes operations, and production deployment automation through a complete end-to-end cloud-native delivery pipeline. [\[SOURCE\]](#) [\[APP\]](#) [\[GITOPS\]](#) [\[IaC\]](#)

Production-Style Multi-Tier Java Application Deployment on Kubernetes

Tech Stack: Kubernetes, kOps, AWS EC2, Docker, NGINX Ingress, Route53, Amazon EBS CSI, MySQL, RabbitMQ, Memcached

Deployed a production-style 5-tier Java application using 8+ interconnected services with persistent storage, ingress, Route53 DNS, and service discovery. Enhanced expertise in Kubernetes networking, storage orchestration, workload scheduling, container networking, and troubleshooting distributed cloud-native applications. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

Production-Grade Cloud Observability Platform on AWS

Tech Stack: Prometheus, Grafana, Loki, Alloy, Flask, AWS, Slack

Implemented centralized metrics, logging, and alerting across 4 AWS EC2 instances using Prometheus, Grafana, Loki, and Alloy. Built production dashboards, PromQL-based monitoring, and Slack notifications while developing practical SRE expertise in observability, incident detection, log analysis, and root-cause investigation. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#) [\[DEMO\]](#)

Serverless AWS Security Compliance Automation Platform

Tech Stack: AWS Lambda, Python (boto3), EventBridge, SNS, S3

Developed a serverless security auditing solution scanning AWS Security Groups across multiple AWS regions, generating compliance reports, storing audit logs in Amazon S3, and delivering automated SNS notifications. Gained hands-on experience in serverless computing, event-driven architectures, Python automation, and AWS security best practices. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

Jenkins-Driven CI/CD Pipeline for Containerized Applications on AWS ECS

Tech Stack: Jenkins, Docker, Maven, SonarQube, Amazon ECS, Amazon ECR

Built an end-to-end CI/CD pipeline integrating 6+ automation stages including Maven builds, SonarQube analysis, Docker image creation, Amazon ECR publishing, and ECS deployments. Enhanced expertise in enterprise CI/CD, container delivery, code quality automation, and release engineering. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#) [\[DEMO\]](#)

Multi-Node Infrastructure Automation using Ansible on AWS

Tech Stack: Ansible, AWS EC2, Linux, SSH, Apache HTTP Server

Automated provisioning and configuration across multiple EC2 instances using Ansible inventory groups, ad-hoc commands, and agentless SSH. Improved deployment consistency while gaining practical experience in configuration management, Linux administration, infrastructure automation, and scalable server provisioning. [\[SOURCE\]](#)

Hardened SRE Monitoring & Security Auditing Agent

Tech Stack: Linux, Bash, Cron, Systemd, OpenSSL, Slack Webhooks

Engineered a production-ready SRE agent that automates Linux system health monitoring, security auditing, log analysis, and real-time ChatOps notifications, improving system reliability through proactive monitoring and operational automation. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

Enterprise DevSecOps CI/CD Pipeline with Automated Security & Quality Gates

Tech Stack: GitHub Actions, Maven, Docker, Trivy, Amazon ECR, Java

Built a production-ready DevSecOps CI/CD pipeline for the vProfile application, automating software delivery with integrated code quality analysis, vulnerability scanning, container image security, and artifact publishing using GitHub Actions, Maven, Trivy, Docker, and Amazon ECR. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

Automated Amazon EKS Infrastructure Provisioning using Terraform

Tech Stack: Terraform, Amazon EKS, AWS VPC, IAM, EC2, Auto Scaling

Provisioned a production-ready Amazon EKS environment using Infrastructure as Code, automating networking, IAM, worker nodes, and Kubernetes cluster deployment. Enhanced expertise in Terraform modules, reusable infrastructure design, cloud networking, and scalable Kubernetes provisioning. [\[SOURCE\]](#) [\[ARCHITECTURE\]](#)

TECHNICAL SKILLS

Programming Languages: Python, Bash, Java, JavaScript, C

Cloud Platforms: Amazon Web Services (AWS), Google Cloud Platform (GCP)

Containers & Orchestration: Docker, Docker Compose, Kubernetes, Amazon EKS, kOps

Infrastructure as Code: Terraform, Ansible, Vagrant

CI/CD, GitOps & DevSecOps: GitHub Actions, GitLab CI/CD, Jenkins, Argo CD, Helm, SonarQube, Trivy

Observability: Prometheus, Grafana, Loki, Alloy, Amazon CloudWatch

AWS Services: EC2, EKS, ECR, ECS, IAM, VPC, Route 53, S3, Lambda, SNS, EventBridge, Elastic Beanstalk

Linux & Networking: Ubuntu, CentOS, Amazon Linux, NGINX, DNS, Load Balancing, MQTT

Databases: MySQL, PostgreSQL, MongoDB

Development: REST APIs, Node.js, Express.js, Laravel, Flutter, n8n

Version Control: Git, GitHub, GitLab, Bitbucket

OTHER SKILLS

Problem Solving	Strong analytical and problem-solving skills with attention to detail
Teamwork	Effective collaboration in cross-functional and team environments
Communication	Professional communication in English and Sinhala
Adaptability	Quick learner with the ability to adapt to new technologies
Time Management	Ability to work independently and manage multiple tasks
Teaching & Mentoring	Experience mentoring and teaching technical concepts

LEADERSHIP, COMMUNITY & TECHNICAL CONTRIBUTIONS

Technical Lead, AWS Student Builder Group, University of Sri Jayewardenepura	2026 – Present
Head of Teaching Panel, FusionX 1.0 AI Bootcamp (SLAAI)	2025
Batch Representative, Faculty of Computing, University of Sri Jayewardenepura	2025 – Present
IEEE Student Branch Member, University of Sri Jayewardenepura	2024 – 2025
Author, Kubernetes Hidden Gems (LinkedIn Technical Newsletter)	2025 – Present
Technical Blogger, Medium	2025 – Present

BLOGS

Getting Started with AWS EKS: A Practical Guide to Managed Kubernetes on AWS
An introductory guide to deploying and managing Kubernetes clusters using Amazon EKS. [\[Read more\]](#)

Model Context Protocol (MCP) Introduction
An introduction to MCP and how AI models securely interact with external tools and services. [\[Read more\]](#)

Kubernetes Hidden Gems #2 — Startup Probes
Exploring Startup Probes for improving application startup reliability in Kubernetes. [\[Read more\]](#)

Kubernetes Hidden Gems #6 — Pod Disruption Budgets (PDBs)
Understanding Pod Disruption Budgets for maintaining application availability during disruptions. [\[Read more\]](#)

[More articles on Medium](#) [Kubernetes Hidden Gems Newsletter \(1K+ Subscribers\)](#)

REFEREES

Shamika Shehan Kosgolle DevOps & SRE Tech Lead, KAYA Kubestronaut (CKS, CKA, CKAD, KCNA, KCSA) Phone: +94 719069583 Email: shamika@kayatech.com	Pubudu S. Jayasekara BSc (Kelaniya), MPhil (Kelaniya) Lecturer, University of Sri Jayewardenepura Phone: +94 711612267 Email: pubudu@sjp.ac.lk
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